

Compiler/program-analysis engineer with professional LLVM 22 and C#.NET static-analysis experience. The public technical lineage starts with compiler/VM projects in 2022, moves through MSIL and native x86-64 in 2023 and AST → IR → x64 in 2024; current proof points are Global-IV, the x86-64 backend, and Roslyn CFG data flow.

2022: Compilers / VMs · 2023: CIL + native x86-64 · 2024–2025: AST / IR / backends → toolchain · 2026 → : Professional LLVM + static analysis

PROFESSIONAL EXPERIENCE	SKILLS
2026 — present ISP RAS — Static Analysis Engineer - Contribute to SharpChecker, ISP RAS's C#.NET static-analysis platform. July — August 2026 MCST — Compiler Engineering Intern - Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety. - Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases. Additional product experience — Diploma.school — client development: jQuery dropdown interactions, responsive mobile/desktop navigation, and UI regression fixes; confirmed no later than January 2025.	Compiler / IR C++23, LLVM 22/IR, APInt, interprocedural analysis, CFG/SSA, dominators, loops, LICM, legality Program Analysis C#.NET, Roslyn ControlFlowGraph, fixed-point data flow, conditional edges, loops/back edges, joins Backend / Codegen x86-64, SysV ABI, NASM IA-32, CDECL/x87, liveness/interference, register allocation, spills, phi lowering Correctness / Tooling LLVM Verifier, conservative rejection, regression checks, differential execution, Linux, CMake
PROJECTS	EDUCATION
LLVM Interprocedural Global IV Optimization An LLVM 22 pass localizes supported induction-like globals after interprocedural effects/affine analysis and a separate legality decision; APInt models fixed-width wraparound. 29 positive/negative regression cases: global-iv,verify, structural expectations, idempotence, and observable before/after execution comparison. x86-64 Compiler Backend & Register Allocation A compiler backend with SSA/CFG validation, liveness/interference analysis, linear-scan and seeded simulated-annealing allocators, spills, phi lowering, and a SysV x86-64 emitter. An independent assignment verifier plus native-vs-interpreter differential execution; tests cover branches, loops, spills, and parallel phi moves; a separate public native-codegen lineage starts with Visk (2023) and Wist4 (2024). DerefAfterNullAnalyzer A Roslyn ControlFlowGraph analyzer propagates null state over conditional edges to a fixed point, handles loops/back edges, and merges states from multiple predecessors. Handles method/property/index/event dereferences and invalidation after assignment/ref/out; dedicated analyzer tests use Microsoft.CodeAnalysis.CSharp.Testing.	HSE University — Software Engineering, 2026–2030 RECOGNITION LangDev'26 — Build the Language, Then Make the Abstractions Disappear Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga. UniversalToolchain — Baltic Science and Engineering Competition First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain. MEPhI Junior Overall winner; First Degree Diploma, 96/100 for UniversalToolchain. Moscow / remote